Granthōmukhīm

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Contents

Chapter 1

The Language

Welcome to this treatise on the Granthōmukhīm language! This is my first conlang which has been in the works for about six years now, with hiatuses in the middle. It started out as an attempt to iron out the irregularities in Pāṇinian Sanskrit (ironic, isn't it?). I took the base vocabulary of Sanskrit and studied the Indo–Aryan and Indo–European branch of languages for information on how all this worked. The very early versions were just Sanskrit words with modified affixes and inflections. It was still Sanskrit, just more Ithkuil–like (I did fall prey to that fallacy). It took a few more years of refinement until the grammar matured enough, so I could wean it off of the ad–hoc Sanskrit lexicon base and introduce a native system. This was also around the same time that I was learning more about the Dravidian languages, as a tangent to my school language courses. The influence of Dravidian is clearly seen in the phonology that developed over time in this language, giving rise to a rich collection of short vowels, retroflexes, liquids and approximants.

While this was in progress, I made an attempt at a highly analytic languge with no inflections or fusional grammar. The entire lexicon was agonstic to the grammar and meaning was achieved by means of particles, prepositions and postpositions. Its vocabulary was not exactly well–researched but more of me rambling and trying to emulate Greek phonology. I did end up using the Greek alphabet for the language. It eventually turned out futile and died out, but what it did teach me was the ease of making an inflectional language with free word–order.

Granthōmukhīm is an Indo–Dravidian language with its lexicon from Indo–Aryan origins, and a highly Dravidian phonology and phonotactics. It has a root–affixing system whereïn atomic base words (called *roots*) are affixed to form *stems* of various classes, which are further injected with nuance and meaning, and finally grammatically inflected before being dropped in a sentence.

Here are the features and characteristics of this Granthōmukhīm:

- Phonemic length, aspiration and vowel quality provide semantic distinction.
- 10 grammatical cases: Nominative, Accusative, Instrumental, Ablative, Dative, Genitive, Inessive, Adessive, Subessive, Vocative
- 3 grammatical genders (which need not be biological and are purely semantic): Masculine, Feminine, Neuter
- 3 grammatical numbers: Singular, Dual, Plural
- A base–10 number system written successively from highest to lowest place, with the units preceding the next lowest place
- Highly agglutinative with morphosyntactic alignment between adjectives, verbs, nouns and pronouns.
- 3 tenses: Present, Past, Future
- 6 aspects: Gnomic, Habitual, Progressive, Aorist, Retrospective, Simple
- 7 moods: Declarative, Conditional, Imperative, Abilitative, Exclamatory, Optative
- Active and passive voices
- Nominal–derived case–based interrogatives and relatives
- 4–way distinction in proximity and in clusivity
- Internal and external sandhi

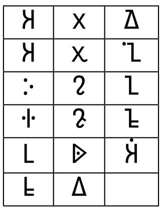
Chapter 2

Writing System & Phonetics

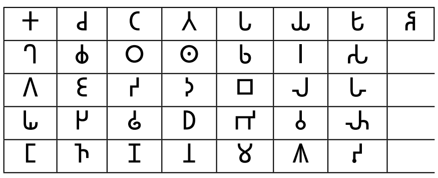
2.1 The Extended Brahmi Abugida

On paper, the language has a native Brahmic script, derived from inspecting and taking inspiration from all the modern Brahmic scripts. It is highly symmetric, with aspiration written as a secondary sound change and possesses regular cononsant clusters in writing. While this is to be written down by hand, there's no Unicode support for a conscript, and hence I needed another script to write it in digital media. Incorporating a digital writing system to complement the spoken standard was a major challenge. In fact, I used Devanagari in the very early stages of the language's development. But I decided that standard Devanagari was lacking and something more substantial was needed to support the language, given its tendency to form long agglutinative words. Given its Indo–Aryan origins, the script ideally should have been a Brahmic abugida. But, for proof of concept, I used an extended version of the modern Hebrew alphabet, repurposing its letters to support the entire phonetic inventory of my language, including vowels, which is not exactly easy, given that Hebrew is an abjad. I made it a partial abugida by requiring every consonant to have an inherent vowel, but wrote down every other vowel beside by manual combination of the aleph and the yod sounds. The geresh was repurposed to cancel the vowel marker. It worked well, although reading it was not exactly easy, for I had no 'training' in reading Hebrew. Eventually, I used the Brahmi alphabet itself, but with a few extensions from Old Tamil. Despite being Indo–Aryan in origin, the phonology of the language is highly influenced by Dravidian, and thus possesses a rich set of liquids and retroflexes, and a lot of short vowels, which the standard Brahmi script does not account for.

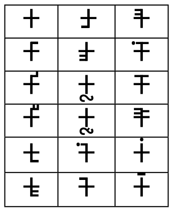
2.1.1 Vowels



2.1.2 Consonants

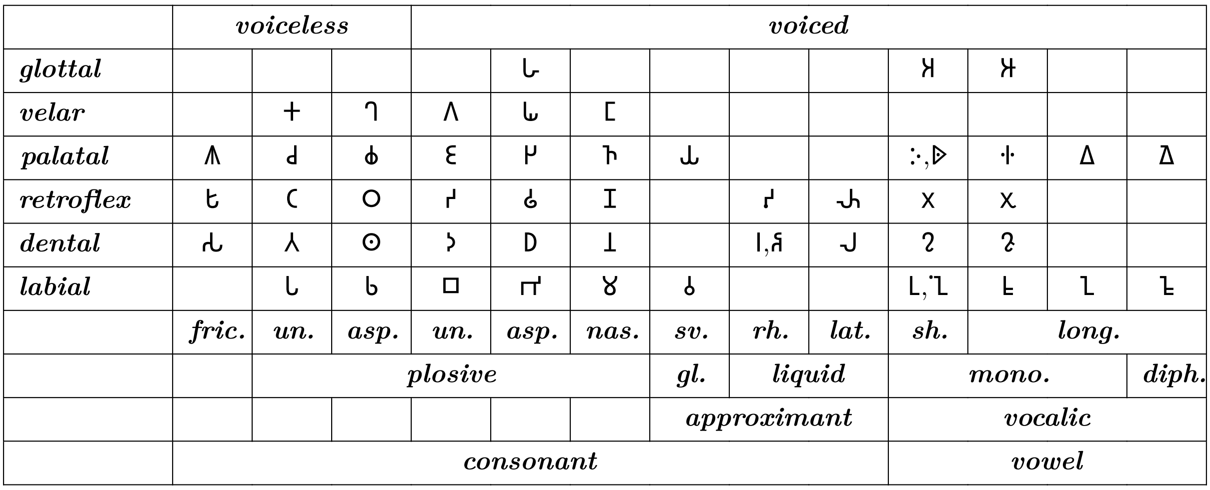


2.1.3 Combinations



2.1.4 IPA Classification

	voiceless			*voiced*									
glottal					ɦ					a	aː		
velar		k	kʰ	g	gʱ	ŋ							
palatal	ʃ	t͡ʃ	t͡ʃʰ	d͡ʒ	d͡ʒʱ	ɲ	j			i, e	iː	eː	ai
retroflex	ʂ	ʈ	ʈʰ	ɖ	ɖʱ	ɳ		ɻ	ɭ	r̩	r̩ː		
dental	s	t̪	t̪ʰ	d̪	d̪ʱ	n		ɾ, r	l	l̩	l̩ː		
labial		p	pʰ	b	bʱ	m	v			u, o	uː	oː	au
	fric.	***un.***	***asp.***	***un.***	***asp.***	***nas.***	***sv.***	***rh.***	***lat.***	***sh.***	***long.***		
		plosive					***gl.***	***liquid***		***mono.***			***diph.***
							approximant			***vocalic***			
	consonant									***vowel***			



- *fric.* — fricative
- *un.* — unaspirated
- *asp.* — aspirated
- *nas.* — nasal
- *sv.* — semivowel
- *rh..* — rhotic
- *lat..* — lateral
- *sh.* — short
- *long.* — long
- *gl..* — glide
- *mono..* — monophthong
- *diph..* — diphthong

2.2 Sandhi

This section specifies all the rules of phonetic conjugation that exist in the language. Any rule that applied to an unaspirated consonant also applies to its aspirated equivalent, though this rule is not necessarily the same for voiced and voiceless counterparts of a sound.

2.2.1 Velars

Antecedent	Succedent	Result
+	Λ	ΛΛ
Λ	+	ΛΛ
+	d	dd
Λ	d	ɛɛ
+□ Λ	ɛ	ɛɛ
+	ƈ	ƈƈ
+	ɼ	ɼɼ
Λ	ƈ, ɼ	ɼɼ
+	ᒐ	ᒐᒐ
+	□	□□
Λ	ᒐ, □	□□
Λ	ɻ	+ɻ
Λ	Ѧ, ƀ	ƀ
+	ɧ, ɼ	+Ɉ
Λ	ɧ, ɼ	ΛɈ
+	ʗ	++
Λ	ʗ	ΛΛ

2.2.2 Palatals

Antecedent	Succedent	Result
d, ɛ	+, Λ	ƀ+
d	ɛ	ɛɛ
ɛ	d	ɛɛ
d	ƈ, ɼ	ƈƈ
ɛ	ƈ, ɼ	ƀƈ
d	ʎ	ʎʎ
d	ʔ	ʔʔ
ɛ	ʎ	ɻʎ
ɛ	ʔ	Ѧʔ
d, ɛ	ᒐ	ƀᒐ
d, ɛ	□	Ѧ□
d	Ѧ, ɻ	dd
ɛ	Ѧ, ɻ	ɛɛ
d, ɛ	ƀ	ƀ
d	Ɉ, ɧ, ɼ	dI
ɛ	Ɉ, ɧ, ɼ	ɛI

2.2.3 Retroflexes

Antecedent	Succedent	Result
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible], [illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible], [illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible], [illegible]	[illegible], [illegible]	[illegible]
[illegible]	[illegible], [illegible], [illegible]	[illegible]
[illegible]	[illegible], [illegible], [illegible]	[illegible]

2.2.4 Dentals

Antecedent	Succedent	Result
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible], [illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible], [illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible], [illegible]	[illegible]	[illegible]
[illegible], [illegible]	[illegible]	[illegible]
[illegible]	[illegible], [illegible], [illegible]	[illegible]
[illegible]	[illegible], [illegible], [illegible]	[illegible]

2.2.5 Bilabial

Antecedent	Succedent	Result
Լ̄	Λ̄	□̄Λ̄
□̄	+̄	□̄Λ̄
Լ̄	Ɛ̄	□̄Ɛ̄
□̄	Ɛ̄	□̄Ɛ̄
Լ̄	ɾ̄	□̄ɾ̄
□̄	Ϲ̄	□̄ɾ̄
Լ̄	ʕ̄	□̄ʕ̄
□̄	λ̄	□̄ʕ̄
Լ̄	□̄	□̄□̄
□̄	Լ̄	□̄□̄
Լ̄	Ɉ̄, Ɉh̄, ɾ̄	Լ̄Ɉ̄
□̄	Ɉ̄, Ɉh̄, ɾ̄	□̄Ɉ̄

2.2.6 Sibilants

The voiceless palato–alveolar sibilant Ʌ̄ mutates into the voiceless retroflex sibilant ᑲ̄ when initial to a velar, retroflex or a bilabial. Hetero–combinations of any of the sibilants (the above two and the voiceless coronal ƛ) results in a voiceless retroflex sibilant geminate.

2.2.7 Voiceled Glottal Fricative/Aspirate

Whenever this sound (Լ̄) is succedent to another velar, palatal, retroflex, dental or bilabial consonant, it leaves the initial alone and mutates into the aspirate (respecting the voice) of the initial.

2.2.8 Nasals

Antecedent	Succedent	Result
[̄, ħ̄, Ī, Ɪ̄	Velar	[̄
[̄, ħ̄, Ī, Ɪ̄	Palatal	ħ̄
[̄, ħ̄, Ī, Ɪ̄	Retroflex	Ī
[̄, ħ̄, Ī, Ɪ̄	Dental	Ɪ̄
[̄, ħ̄, Ī, Ɪ̄	Bilabial	Ȣ̄
Velar	Ȣ̄	[̄Ȣ̄
Palatal	Ȣ̄	ħ̄Ȣ̄
Retroflex	Ȣ̄	ĪȢ̄
Dental	Ȣ̄	Ɪ̄Ȣ̄
Bilabial	Ȣ̄	Ȣ̄Ȣ̄
Non–Labial Nasal 1	Non–Labial Nasal 2	Nasal 1 geminate

2.2.9 Vowels

- Any vowel encountering itself (even with different length) becomes a long vowel.
- The next set of rules combine vowel classes when the initial is a monophthong.
  - Ӽ + ∵ — ▷
  - Ӽ + L — ̇L
  - Ӽ + ·ǀ· or Ӿ + ∵ or Ӿ + ·ǀ· — Δ
  - Ӽ + L or Ӿ + L or Ӿ + Ŀ — Ɩ
  - Ӽ, Ӿ + ▷, Δ — Δ
  - Ӽ, Ӿ + ̇L, Ɩ — Ŀ

- ⁖ + ꓘ, ꓘ̵ — ᒍ, ᒍ̵
- ·ǀ· + ꓘ, ꓘ̵ — ·ǀ·ᒍ, ·ǀ·ᒍ̵
- ⁖ + L, Ŀ — ᒍ̦, ᒍ̦̦
- ·ǀ· + L, Ŀ — ·ǀ·ᒍ̦, ·ǀ·ᒍ̦̦
- ⁖ + ᐅ, Δ — ˙ᒍ, ᒍ
- ·ǀ· + ᐅ, Δ — ·ǀ·˙ᒍ, ·ǀ·ᒍ
- ⁖ + ˙ʅ, ʅ — ˙ᒍ̄, ᒍ̄
- ·ǀ· + ˙ʅ, ʅ — ·ǀ·˙ᒍ̄, ·ǀ·ᒍ̄
- L + ꓘ, ꓘ̵ — ȣ, δ
- Ŀ + ꓘ, ꓘ̵ — Ŀȣ, Ŀδ
- L + ⁖, ·ǀ· — δ́, δ̋
- Ŀ + ⁖, ·ǀ· — Ŀδ́, Ŀδ̋
- L + ᐅ, Δ — ˙ʓ, ʓ
- Ŀ + ᐅ, Δ — Ŀ˙ʓ, Ŀʓ
- L + ˙ʅ, ʅ — ˙ʓ̄, ʓ̄
- Ŀ + ˙ʅ, ʅ — Ŀ˙ʓ̄, Ŀʓ̄

- The next set of rules are for when the initial is a diphthong (either regular or monophthongised).
  - If the initial is a palatal, then it is left as it is and the final vowel is attached to ᒍ̄ (respecting length)
  - If the initial is a labial, then it is left as it is and the final vowel is attached to ȣ̄ (respecting length)
  - In the above two cases, the resultant is long if and only if at least one of the two vowels is long.

2.3 Phonotactics

2.3.1 External Sandhi

The previous section detailing the rules of phonetic conjunction are usually applicable to external sandhi. Roots (see §*4.5*) native to the language's vocabulary may have 'illegal' phonetic combinations without merging. This is attributed to an internal effort to keep at bay too many unrelated homophones. However, in accordance with natural speech, the phonetic boundaries at the extremes of words may merge in accordance with external sandhi rules.

2.3.2 Internal Sandhi

It is worth noting that the word *sandhi* is being used very loosely in this section, for it not only encompasses phonetic conjunction, but also phonetic mutation under certain phonetic environments. These rules apply primarily to root words and affixes used in the formation of words, although complete words may be subject to this if applicable.
It is also worth knowing that these rules are not exactly 'regular', in the sense that every possible case cannot be covered by a set of axioms, and intuition and experience will have to play a bigger role, for some instances of these rules may seem arbitrary.

2.3.2.1 Nasal Retroflexion

This sound change is very much at home, given the linguistic family that this language is set in. The cononsant ꓕ may mutate into the retroflex Ɪ when it is:

- Preceded by a velar and, a long vowel or a (regular/monophthongised) diphthong.

- Preceded by a semivowel or an alveolar flap/trill or a dental lateral approximant, and a long vowel or a (regular/monophthongised) diphthong.
- Preceded by the voiceless retroflex sibilant (ƅ).
- Succeeded by the long vowel Ӿ in its syllable and the lateral retroflex approximant (ɺ).

2.3.2.2 Geminate Trill

The denti–alveolar flap (ǀ) becomes a denti–alveolar trill (Ꞅ) when it occurs as a geminate.

2.3.2.3 Lateral Retroflexion

This phenomenon occurs predominantly in the Dravidian languages. The dental lateral approximant (ᒍ) becomes the lateral retroflex approximant (ɺ) when it is:

- Preceded by a velar with the vowel Ӿ, Δ or ʟ.
- Preceded by a palatal with the vowel ʟ.
- Preceded by a dental with the vowel Ӽ or Ӿ.
- Preceded by a bilabial or an alveolar flap/trill with the vowel Ӿ.
- Preceded by the voiceless retroflex sibilant ƅ with the vowel ʟ.

2.3.2.4 Lateral to Rhotic Decay

This interesting phenomenon sometimes occurs in Tamil and Malayalam. When a lateral approximant (ᒍ, ɺ) is followed in a consonant cluster by a velar, it mutates into a rhotic retroflex (ɽ).

2.3.2.5 Anti–Retroflexion

Due to the above rules or otherwise, if certain retroflex syllables occur side by side, then the first retroflex consonant mutates into a non–retroflex version of itself. I becomes a ⊥ when followed by any of the retroflex *varga* members or an alveolar tap/trill or a ɺ or ɽ. Similarly, when followed by the same consonants, a ɺ becomes a ᒍ.

Chapter 3

Grammar

3.1 Nominals

Given its Indo–Aryan origins, the grammar specifies an extended system of nine true nominal grammatical cases — nominative (default), accusative, instrumental, ablative, dative, inessive, adessive, subessive and vocative — and one pseudo–nominal adjectival genitive case. Unlike other Indo–Aryan languages, the vocative is treated as a true nominal case and not as a subsidiary nominative. The grammar allows for interesting recursive and paradoxical vocative usages. The system is fully nominative–accusative, unlike some ergative–absolutive Indo–Aryan descendants. There are three grammatical genders and three cardinal and ordinal numbers.

3.1.1 Nouns

Given in table 3.1 is a list of declensions taken by three words Γɤ, Jƛ and ȯ⊥̇, which are a boy's name, a girl's name, and a forest, in the masculine, the feminine and the neuter, respectively.

Case	Masculine	Feminine	Neuter
Nominative	Γɤ	Jƛ	ȯ⊥̇
Accusative	Γɤ̇	Jƛ̇	ȯ⊥ɤ⊥̇
Instrumental	Γ⁊ɤI	JλƗ	ȯ⊥ɤ⊥⁊ɤI
Ablative	Γɤ̄	Jƛǂ	ȯ⊥ɤ⊥ɤ̄
Dative	Γɤ⫫	Jƛ⫫	ȯ⊥ɤ⊥⫫
Genitive	Γɤʔ	Jƛʔ	ȯ⊥ɤ⊥ʔ
Inessive	Γ̈ɤ	Jλ	ȯ⊥ɤ˥
Adessive	ΓɤJh	JƛJh	ȯ⊥ɤ⊥Jh
Subessive	Γɤ·Ŧ	JƛŦ	ȯ⊥ɤ⊥Ŧ
Vocative	Γ⁊ɤ	Jƛ⁊ɤ	ȯ⊥ɤ⊥⁊ɤ

Table 3.1: Noun Declensions

3.1.2 Pronouns

Given in table 3.2 is a list of all the pronouns, of which three classes — personal, relative and interrogative.

Case	**First**	**Second**	**Third**	**Relative**	**Interrogative**
Nominative	ɤ́	ɾ́	λ́	ɺ́	†́
Accusative	ɤ̇◌́	ɾ̇	λ̇	ɺ̇	†̇
Instrumental	ɤ̿	ɾ̿	λ̿	ɺ̿	†̿
Ablative	ɤ̄I	ɾ̄I	λ̄I	ɺ̄I	†̄I
Dative	ɤɺ	ɾɺ	λɺ	ɺɺ	†ɺ
Genitive	ɤʔ	ɾʔ	λʔ	ɺʔ	†ʔ
Inessive	ɤ̄	ɾ̄	λ̄	ɺ̄	†̄
Adessive	ɤɟ	ɾɟ	λɟ	ɺɟ	†ɟ
Subessive	ɤĪ	ɾĪ	λĪ	ɺĪ	†Ī
Vocative	ɤʅ	ɾʅ	λʅ	ɺʅ	†ʅ

Table 3.2: Pronouns

The three genders are intrinsic to a noun, but not to a pronoun and thus need to be suffixed separately, as specified in table 3.3.

Gender	**Suffix**
Masculine	(default)
Feminine	ʅ
Neuter	ʅ̇

Table 3.3: Grammatical Gender Suffixes

Both nouns and pronouns are inflected for three numbers in two forms as shown in table 3.4.

Type	**Singular**	**Dual**	**Plural**
Cardinal	(default)	ʗ	ð
Ordinal	ɺ	ʖ	λ

Table 3.4: Number Suffixes

3.1.3 Declining Nominals

We have the following structure for declining a noun: *noun–root* + *case* + *number*, and for declining a pronoun: *pronoun–root* + *case* + *gender* + *number*. Note that inflecting for anything but grammatical case is done only once, when the topic of the sentence is introduced, or a new topic is injected. In future instances, unless deemed necessary, context is sufficient and the other inflections can be omitted. Here are a few examples to illustrate:

- ðɪɤɭʗ — in two/a pair of forests
- †̄Iʅð — with whom *(feminine plural)*?

3.1.4 Numbers

Numbers by themselves are treated as pseudo–adjectives. They are inflected as: *number–root* + *gender* + *nature (cardinal/ordinal)*. The suffixes are identical. Table 3.5 lists out all the numbers in existence.

0	[script]	10	[script]	40	[script]
1	[script]	11	[script]	50	[script]
2	[script]	12	[script]	90	[script]
3	[script]	13	[script]	100	[script]
4	[script]	19	[script]	200	[script]
5	[script]	20	[script]	900	[script]
6	[script]	21	[script]	1000	[script]
7	[script]	22	[script]	10^4	[script]
8	[script]	29	[script]	10^5	[script]
9	[script]	30	[script]	10^7	[script]

Table 3.5: Numbers

For example: [script] — 12345 *(masculine)*

3.1.5 Adjectives

Adjectives are inflected for gender and number, and in a noun phrase, they always appear before the noun that they are qualifying. Although the word order is free in a sentence, this relative order is enforced within a noun phrase.

For example: [script] [script] is a *good boy*, but [script] [script] *is a good girl*.

3.1.6 Adverbs

Adverbs are not well–defined in the English sense, and they are treated as locative adjectives. They are simply bare adjective roots suffixed with [script] and are not inflected for number and gender.

For example: [script] means *happily*

3.2 Verbs

Verbs are perhaps the densest elements of grammar in this specification. They are highly agglutinative and inflected for several dimensions of agreement. Inflections also help express deep nuance in concepts of time and mood. The general inflection structure of a verb is as follows:

optional negation + verb–root + person + gender + number + tense & aspect + mood

It is important to note that gender, number and person are inflected with respect to the doër. The gender suffixes are identical to the ones above. Negation (if needed) is expressed by a simple prefix [script].

The grammar specifies ten tenses, and their suffix markers are given in table 3.6. It is worth noting that some tenses are mutable, and are subject to interpretation under context.

Tense	Aspect	Suffix
Present	Gnomic	(default)
Present	Habitual	[script]
Present	Progressive	[script]
Past	Aorist	[script]
Past	Progressive	[script]
Past	Retrospective	[script]
Past	Habitual	[script]
Future	Simple	[script]
Future	Progressive	[script]
Future	Retrospective	[script]

Table 3.6: Tense Suffixes

There are seven moods, which are used to increase the nuance expressed via a verb. Important grammatical functions such as conditionals and abilitatives are considered aspects of a verb. Note that an aspect cannot stand by itself without a tense. The suffixes are listed in table 3.7. Compound moods are subsidiary and may be used as shown:

Mood	**Suffix**
Declarative	(default)
Conditional	ꝺ
Imperative	ᒷ
Interrogative	ϯ
Abilitative	⟂
Exclamatory	ᒣ
Optative	ɤ
Conditional–Interrogative	λ
Conditional–Abilitative	J
Conditional–Optative	ᒪλ
Abilitative–Interrogative	Ɪ
Abilitative–Exclamatory	Ɪᒣ
Optative–Interrogative	ɤϯ
Optative–Exclamatory	ɤᒣ

Table 3.7: Mood Suffixes

Table 3.8 lists the suffixes used to mark person.

Person	**Suffix**
First	ɤʼ
Second	ᒪ
Third	(default)

Table 3.8: Person Markers

Here's an example to illustrate all the moving parts in action:
⟂ϯГᒣᒣꝺJГ⟂ — *they (feminine) couldn't have played*

3.2.1 The Periphrastic Passive

So far, the verb structure used is purely in the active voice, which is usually the default tone. However, when necessary, the grammar specifies a passive voice, which can be formed in two ways. One (and the most common) way is by mutating the verb root itself into a passive form and then using the inflections specified by the active form, or (less commonly) by using the following structure:

instrumental subject + past form of the verb in agreement

Note that this method of forming the passive is only defined for the present tense, and does not exist for the past and the future tenses. For those, the verb root must be mutated. The past *form* means that the perfective aspect is carried over, as in, simple to simple and progressive to progressive. The verb in this new form must be inflected to agree with the subject in the instrumental form that comes with it. In this verb phrase, the order can be switched around, but the two components move together as a unit and cannot be separated. Such conveniences are only facilitated by mutating the verb root. When the subject is unspecified, use the neuter singular.

For example: λƗλ ꝺꝺᒪГϯ — it has been told

3.3 Determiners & Conjunctives

3.3.1 Conjunctions

These are simple atoms that appear in between connecting clauses. Table 3.9 lists all the conjunctions.

and	ᑯ
or	ȯ
but	+̰
enough/so/even	ᒣ̄ȯ
than	Ɪ̄

Table 3.9: Conjunctions

3.3.2 Interrogatives & Relatives

For nuanced interrogative/relative words, they are formed by taking the inflected forms of the interrogative/relative pronouns and mutating their case inflections as given in table 3.10, but keeping the other inflections for number and gender the same. For most words, unless needed, the other inflections for number and gender can be dropped.

Word	**Case**	**Intended Meaning**
Why	Accusative	for what
How	Instrumental	by what
When	Dative	at what
Where	Locative (or its subcases)	in what

Table 3.10: Nuanced Interrogative/Relative Words

3.3.3 Proximity

When a third person personal pronoun is used with all of its inflections, it fails to specify the proximity of the specified with respect to the speaker and the listener. Table 3.11 lists the prefixes used to specify that.

Prefix	**Proximity**
(default)	unspecific/away from both the speaker and the listener
ᒪ̄	away from the speaker and close to the listener
˙ᒪ	close to the speaker, away from the listener
ᒪ	close to both the speaker and the listener

Table 3.11: Proximity Markers

3.3.4 Clusivity

Because of three numbers, clusivity is far more intricate than usual. It is also marked by a prefix to a first person dual or plural personal pronoun. Table 3.12 lists all the clusivity markers.

Prefix	**Meaning**	**Clusivity**
(default)	speaker and direct listener; excluding passive listener	Dual Inclusive
˙ᒪ	speaker and passive listener; excluding direct listener	Dual Exclusive
˙ᒪ	including direct listener	Plural Inclusive
(default)	excluding direct listener	Plural Exclusive

Table 3.12: Clusivity Markers

Chapter 4

Lexicon

4.1 Conception & The Root System

While scouring the universe for various grammatical lexicon and word–building systems, I looked through various forms and practices in languages across the spectrum. Two systems, in particular, caught my eye, the Sanskrit–Dravidian root affixation system and the Semitic multi–consonantal template system. The Semitic system seemed very logical and straightforward. Although it was originally a similar affixing system just like the Indo–European equivalent, millenia of elision and evolution has turned it into this logical looking template system. A bit more research showed me that having an unnaturally obtrusive template system is not really good (obviously for the naturality) of the language, and so I settled for my native system.

Take the root 𑀓𑀾 in Sanskrit, which is associated with the action of *doing*. From it, by attaching various affixes (and millenia of sandhi and evolution), we get words like 𑀘𑀓𑁆𑀭 for *wheel*, 𑀓𑀭𑁆𑀫 for *action* 𑀓𑀸𑀭𑀡 for *reason*, and 𑀓𑀭𑁆𑀢𑀸 for *doër*. This language specifies a very similar, yet highly regular affixing system for taking atomic roots and converting them to grammatically sound and nuanced stems.

4.2 Derivation

Using the atomic bare roots as a base, there are many ways of affixing (subject to *sandhi*) it to deepen its nuance, mutate its meaning, or assign different grammatical functions to it. Every single nominal and verb in existence in the language is essentially an inflected affixed root stem. We will study the affix system by categorising affixes on the basis of grammatical function. Each section will begin with an affixing rule and a few examples after.

The affix system is divided into two classes: strong and weak, or class I and II, respectively. In a final grammatically valid sentence, there can only exist strong forms of any root. A strong form of a root is sort of the final form of the root, with all its nuances and embellishments injected, ready for either standing by itself or being inflected grammatically rather than semantically. A weak form of a root is an incomplete variant of the root, and needs to be semantically inflected further. It cannot stand by itself in a sentence and no grammatical affixes can be attached to a weak form directly. A class I affix produces a strong form, whereas a class II affix produces a weak form. In other words, a strong form is a stem, whereas a weak form is a root.

4.2.1 Infinitive (class I)

Rule: *root* + 𑀅𑀢𑀼𑀁

- 𑀓𑀢𑀼𑀁 — to do
- 𑀤𑀢𑀼𑀁 — to see
- 𑀫𑀢𑀼𑀁 — to die
- 𑀰𑀢𑀼𑀁 — to purify
- 𑀣𑀺𑀭𑀼𑀁 — to solidify/freeze
- 𑀪𑀸𑀢𑀼𑀁 — to shine

4.2.2 Negation (class I & II)

Rule: ꓘI̅ + *root*

- ꓘI̅ꞮJ̄ — wake up
- ꓘI̅ꞩꓥ̄ — slow
- ꓘꞱꞍ̄ԁ — empty
- ꓘȣ̄ȣ̄ʖ̄ — sad

4.2.3 Reduplication (class II)

Rule: Double the the first syllable of the input root to increase emphasis/intensify its meaning.

- ꝉꝉλ̄ — toil
- ԼԼʌ̄ — stare
- ꓘȣ̄ȣ̄ʖ̄ — agony
- ʌʌ̇ — wander

4.2.4 Nominalisation (class I)

Rule:

- *root* + ꓘλ + *short vowel (masculine) / long vowel (feminine) /* ꓘ̇ *(neuter)*
- The masculine and feminine forms make it an adjective inflected for gender, while the neuter form makes it a noun.

- Լʌλ̇ — vision
- ꓘ̇Լʌλ — blind (masculine)
- ȣλλ̇ — death
- ȣλλ̄ — dead (feminine)
- ꝉλλ̇ — action

4.2.5 Causative Infinitive (class I)

Rule: *root* + ꓘԼ̄I

- ԼʌԼ̄I — to show (*lit.* to cause to see)
- ȣλԼ̄I — to kill (*lit.* to cause to die)
- ꞙI̅λԼ̄I — to polish (*lit.* to cause to shine)
- ꝺʖԼ̄I — to teach (*lit.* to cause knowledge)

4.2.6 Active (class II)

Rule: *root* + ꓘ Ŧ

- ꞆJŦλ — king (*lit.* one who rules)
- ꞆJŦλ̇ — government (*lit.* one which rules)
- ꝉꝉŦλ — artist (*lit.* one who paints)
- ꝉꝉŦλ̇ — brush (*lit.* one which paints)

4.2.7 Passive (class II)

Rule: *root* + ꓘԼ̄ꞁ

- ԼJԼ̄ꞁλ̇ — kingdom (*lit.* one which is ruled)
- ꝺʖԼ̄ꞁλ̇ — speech (*lit.* one which is said)

4.2.8 Compounds (class II)

Rule: *root* + *root* (subject to sandhi)

- ꞁε̄εJλ̇ — ocean (*lit.* big water: ꞁʖ̄ + εJ̄)
- ꓘ̇ꞁε̄εJλ̇ — pond (*lit.* small water: ꓘI̅ + ꞁʖ̄ + εJ̄)
- ꞅꞅꞆꞙI̅λ̄ꞆʖŦλ̇ — plant (*lit.* sunlight – one which eats: ꞅꞅI̅ + ꞙI̅λ̄ + Ꞇʖ̄)

4.3 Colours

Colours are derived from basic elements of nature. To darken a shade, prefix the particle , and negate it to lighten the shade.

Colour	Word	Root
Black		Darkness
White		Milk
Red		Blood
Green		Plant
Blue		Sky
Yellow		Gold
Orange		Sun
Brown		Bark
Violet/Purple		Grape

A few derived colours are:

- Grey — (*lit.* light black)
- Grey — (*lit.* dark white)
- Pink — (*lit.* light red)
- Vermillion — (*lit.* dark orange)

4.4 Sample Text

This is the first clause of the Universal Declaration of Human Rights:

> *All human beings are born free and equal in dignity and rights. They are endowed with reason and conscience and should act towards one another in a spirit of brotherhood.*

It is translated as follows:

‖ ‖

4.5 Native Roots

The following glossary gives a list of roots used to build the lexicon. In each entry, the left word in bold is the root. The right word is either a root, or a pointer to several compounding entires in the same glossary, which can be compounded according to the rules specified in the grammar (§*4.2.8*).

A

abandon
ability
above
absorb
accept
ache
acid
act
action
active
acute
adapt
add
adhesion
adjust *adapt*
adult *big*, *man*
after
again
age
agree *accept*, *moral*
air
alive
all
allow
almost
alone
altitude
always
amorphous
amount
amuse
anchor
anger
angle
animal
ankle

answer
ant
anxiety
any
apart
apex *top*
appear
apply
approach
arch
area
argue
arm
army
around
arrange
arrive *come*
art
ash
ask
attack
attempt
attract
authority *right*
autumn
awake *sleep*
awe *wonder*
axis

B

baby *immediate*, *life*
back
bad *moral*
bag *hold*
bake *cook*
ball *round*, *play*
band
bare
bark
base
bat
bath
be
bear *fear*, *hair*, *animal*
beat
beauty
bed *sleep*, *thing*
bee *honey*, *create*
before *after*
begin
behind
believe
bell
belly
below *above*
bend
berry *fruit*, *size*
between *waist*
big *size*
bind
bird
birth *life*
bite
bitter
blade
blame
blank
blind *vision*
blood
blow
board
boat
body
boil
bone
book *knowledge*, *give*
border
borrow *loan*
bottom *top*
bow
bowl *food*, *hold*
box *thing*, *hold*
boy *man*, *size*
brain
branch
brass
brave
bread
break
breast *milk*, *create*
breath *man*, *air*
brick
bridge
bright *light*
bring *come*
broad *wide*
brother
brush *colour*, *tail*
build
burn
burst *explode*
bury
bush
busy
buy *receive*, *monetary*